

InstructBLIP: Towards General-purpose Vision-Language Models with Instruction Tuning

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- Introduction
- Vision-Language Instruction Tuning
- Results and Analysis
- Related Work
- Conclusion

BackGround

Vision-Language
Models

General-Purpose
Vision-Language
Model

Instruction
Tuning

LLM

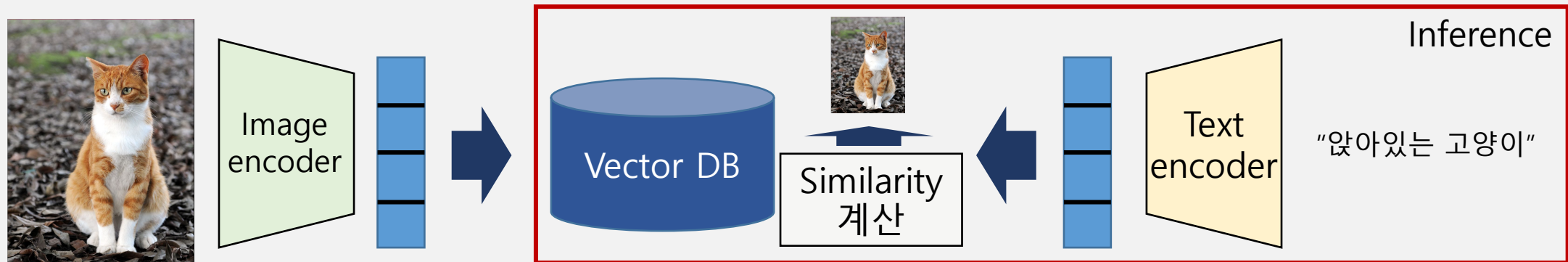
Background Vision-Language Model

- Vision-Language Model이란 무엇인가?
Image와 Text 두 가지를 모두 처리할 수 있는 모델

- Examples of tasks

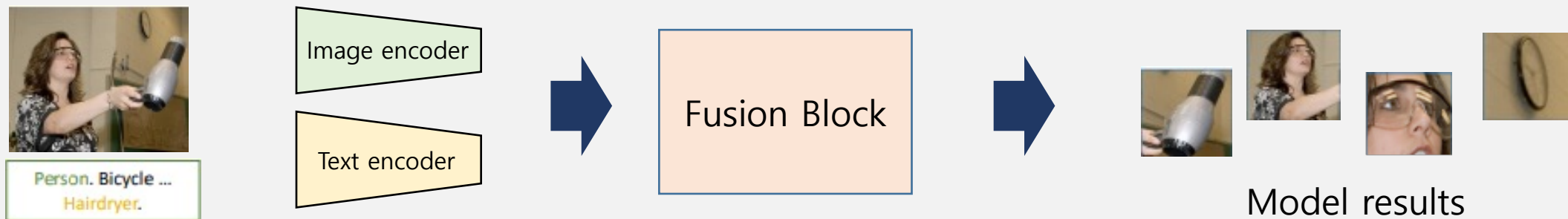
- Image retrieval from natural language text

➡ 텍스트와 가장 유사한 이미지를 뽑는 Task 사전에 DB에 저장해두고 Text와 유사한 이미지 추출



- Phrase grounding, i.e., performing object detection from an input image and natural language phrase

➡ 이미지와 텍스트가 들어오면 이미지 내에서 텍스트와 관련된 Object를 찾아내는 Task

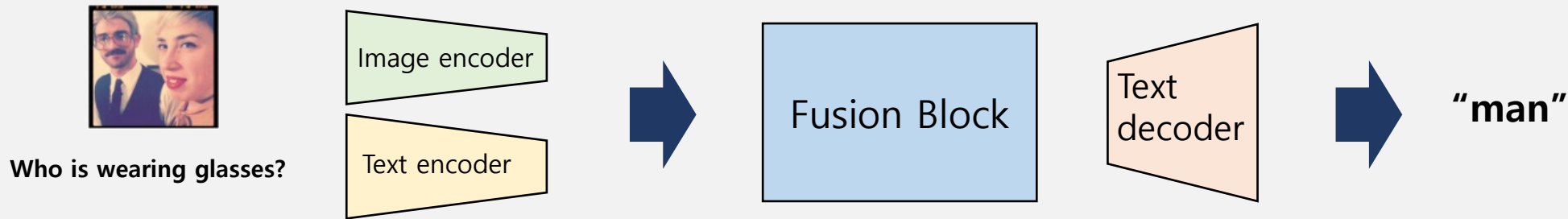


Background Vision-Language Model

- Examples of tasks

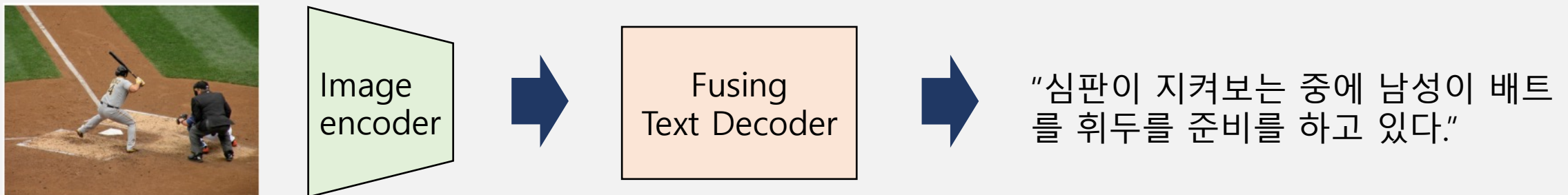
- Visual question answering, i.e., finding answers from an input image and a question in natural language

➡ 이미지를 바탕으로 질문을 했을 때 그에 대한 답변을 생성하는 Task



- Generate a caption for a given image. This can also take the form of conditional text generation, where you'd start with a natural language prompt and an image

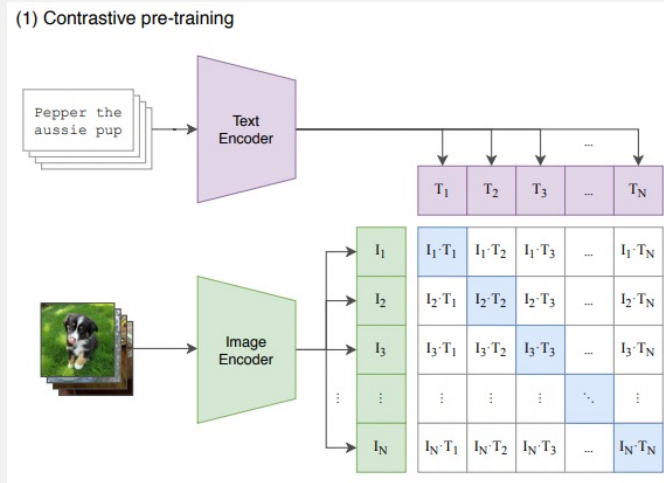
➡ 이미지에 어울리는 캡션을 달아주는 Task



Background Vision-Language Model

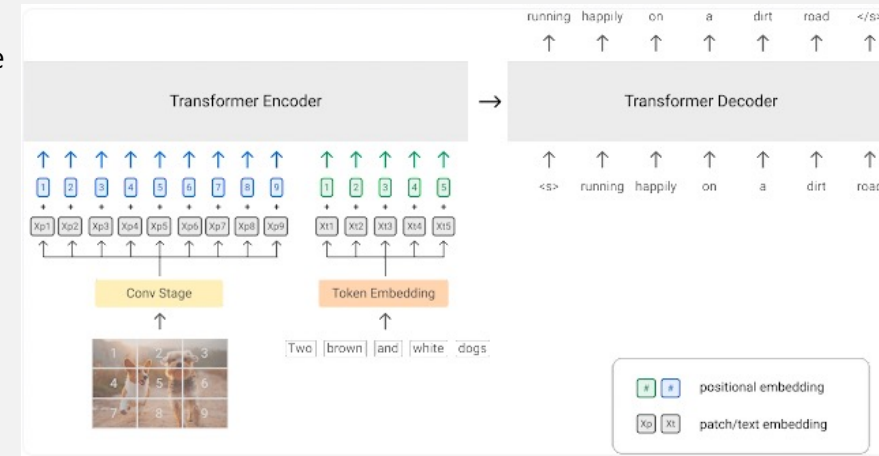
- Training Strategies of Vision-Language Models

• Contrastive Learning



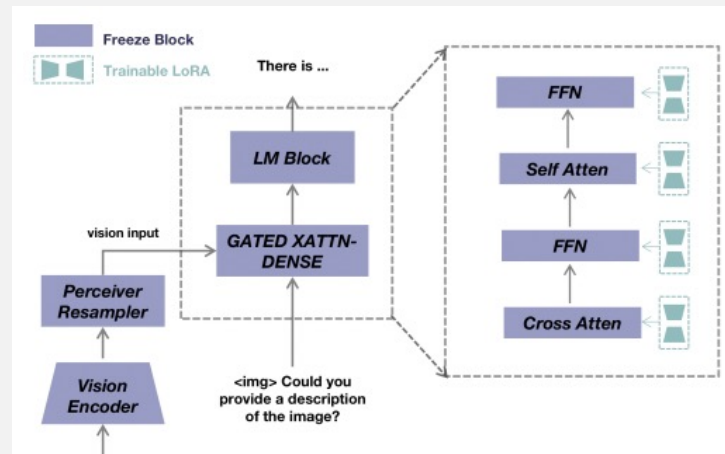
- 이미지와, 텍스트의 feature space를 같은 벡터 공간으로 매핑
- Pair가 답인 경우에 similarity를 1, 다른 경우에 similarity를 0으로 학습

• Prefix Language Modeling



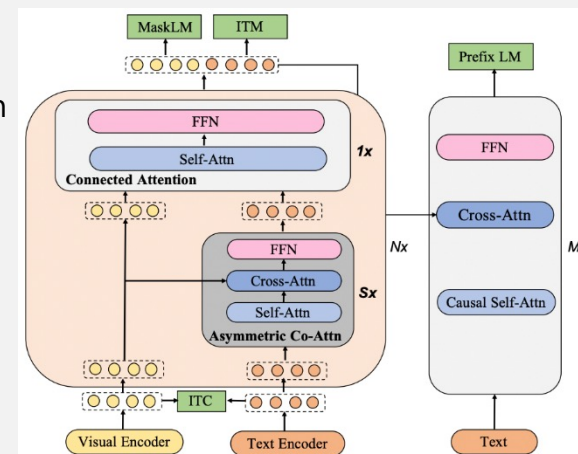
- Decoder 모델의 생성 : 앞에서 이미지에 대한 처리와 문장의 시작이 되는 텍스트를 넣고 이후로 Decoder를 활용해 생성

• Multi-modal Fusing with Cross Attention



- Cross Attention : Visual Feature를 Attention의 key, value 값에 넣고 Query에 text feature를 넣음으로써 이미지와 텍스트 Feature를 Fusing

• Masked-Language Modeling / Image-Text matching



- Masked-Language Modeling (MLM) : Bert의 Pre-training 기법과 동일하며 Text의 $n\%$ 를 마스크로 두고 예측하도록 학습
- Image-Text matching (ITM) 최종 joint-representation 임베딩을 바탕으로 텍스트와 이미지가 일치하는지 확인

Background **General-Purpose vision-language Model**

- 다양한 Vision-Language Task를 하나의 모델로 한번에 할 수 있을까??

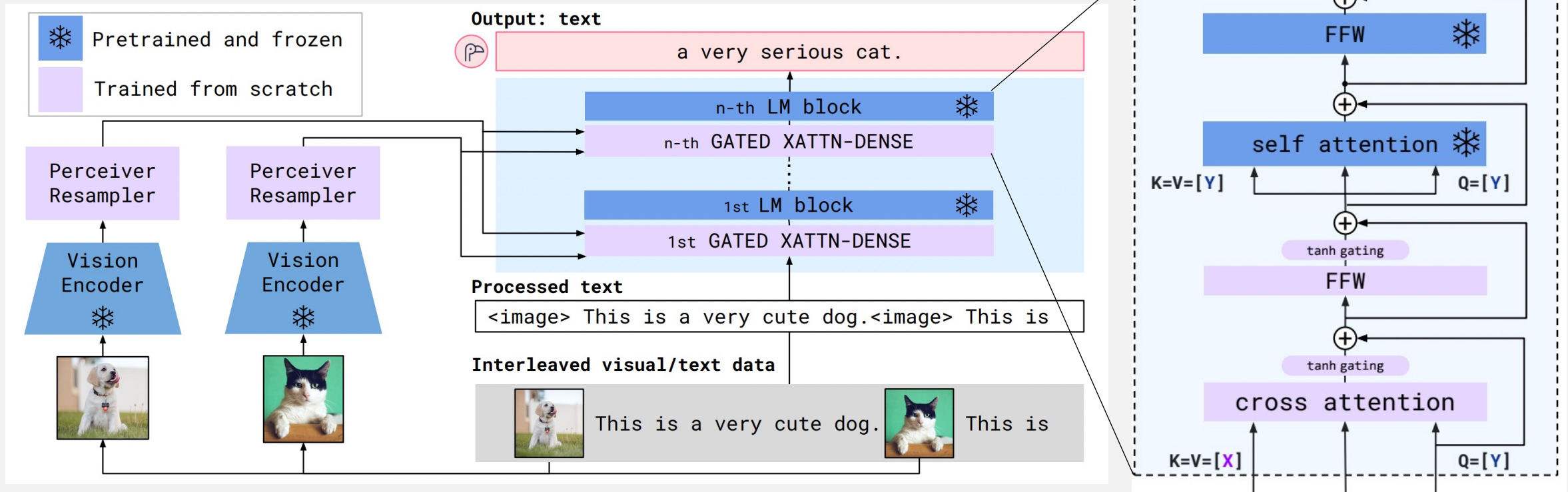
$$p(x) = \prod_{i=1}^n p(s_i | s_1, \dots, s_{i-1})$$

$$p(output | input, task)$$

- 초거대 언어 모델과 연결하여 다양한 Vision-Language Task를 가능하도록 구현하는 연구흐름이 생김
 - DeepMind 의 Flamingo
 - Salesforce 의 BLIP-2

Background **General-Purpose vision-language Model**

- Flamingo



- Framework

Pre-trained image encoder

: freezed NFNet

Pre-trained text decoder

: feezed Chinchilla

- Perceiver Resampler

Vision-Encoder를 바탕으로 생성된

Visual feature를 visual token으로 매핑

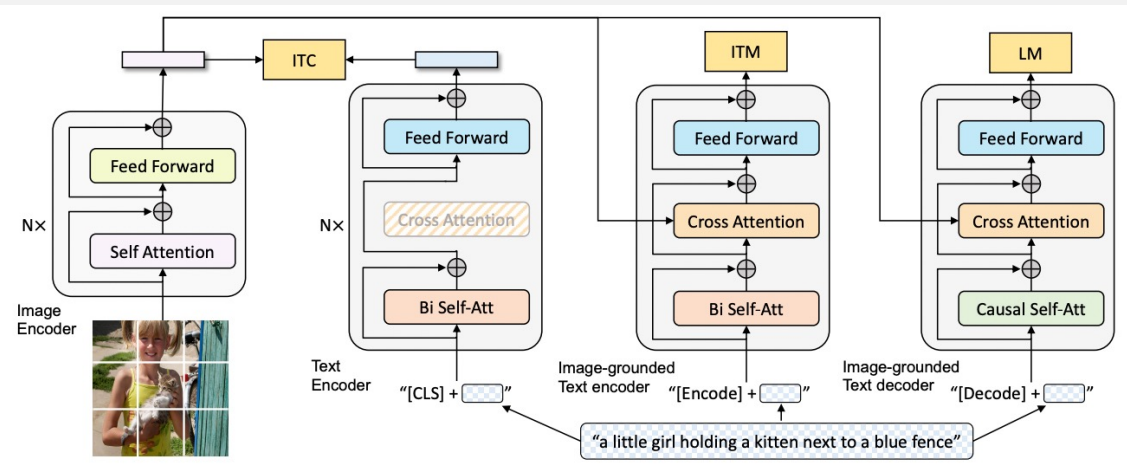
- Gated Cross Attention Layer

Key, Value of attention : visual Feature

Query of attention : language Feature

Background General-Purpose vision-language Model

- BLIP



BLIP : 데이터 관점에서의 개선과 Objective Function 관점에서의 개선

- * 데이터 관점에서의 개선

web data에 노이즈가 많은 점을 문제로 정의하고
noise 데이터를 제거하는 프로세스를 통해 data cleaning

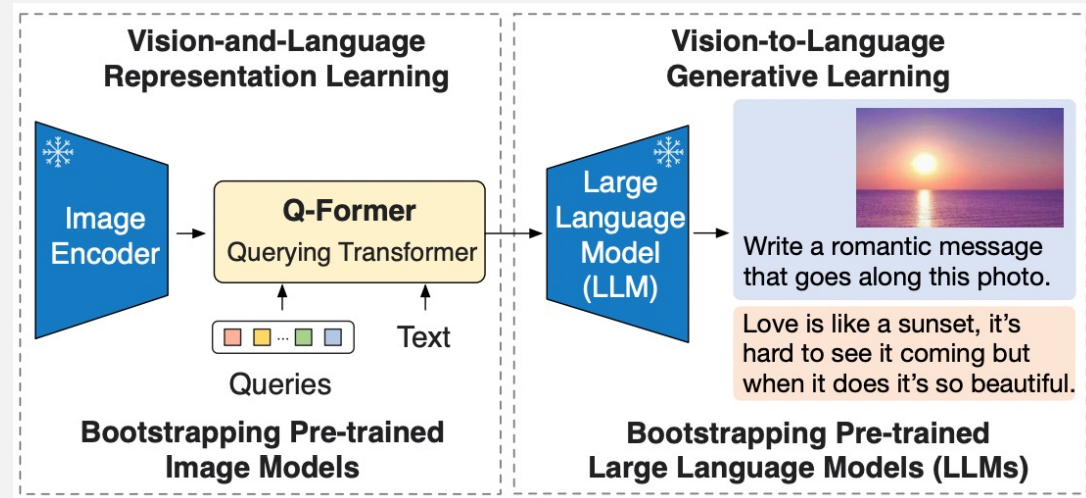
- * Objective Function 관점에서의 개선

Image-text contrastive(ITC) Loss

Image-text matching (ITM) Loss

Language Modeling (LM) Loss

- BLIP-2



BLIP-2 : Image, Text Modality 사이의 차이를 줄이는 관점으로 접근

- * Q-former를 활용하여 modality gap 을 줄이고자 함

- * Step 1 : Freeze Image encoder pre-training

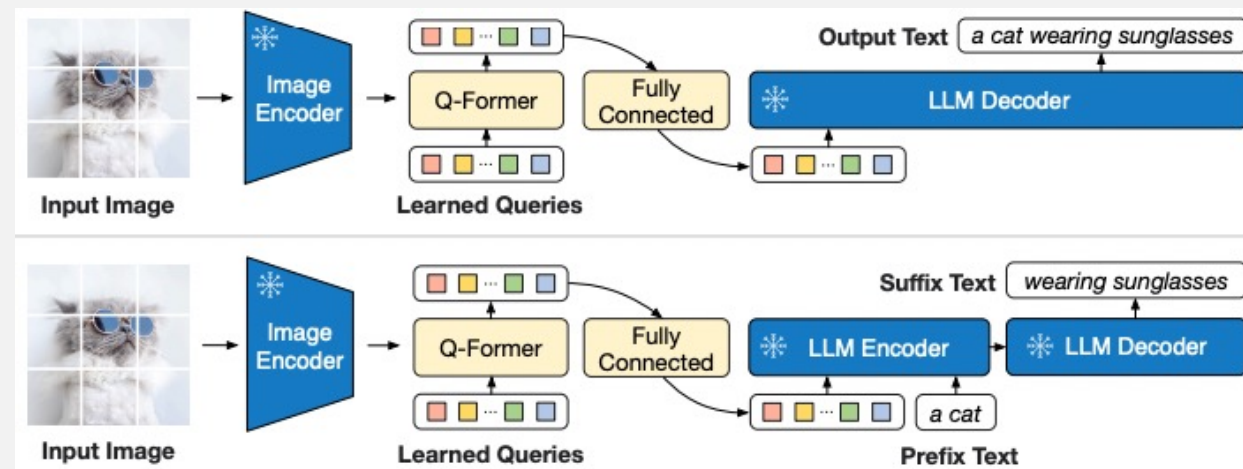
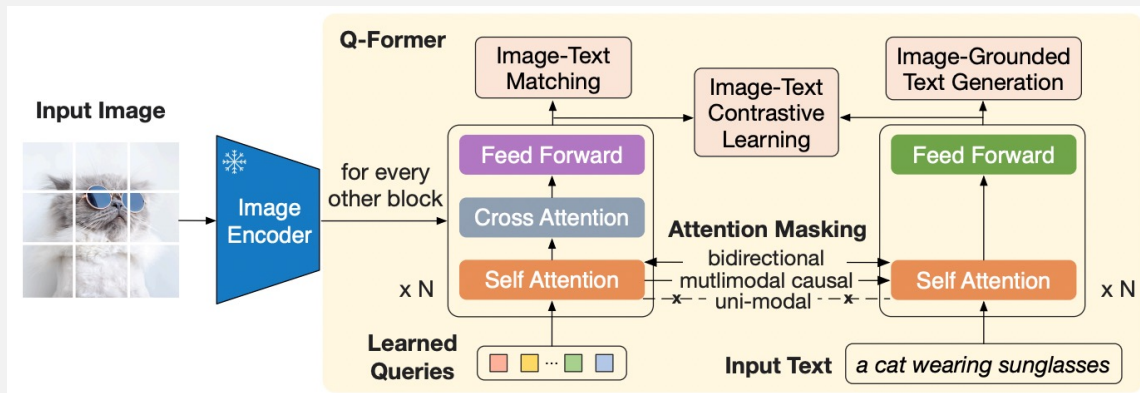
Image-text contrastive(ITC), Image-text matching(ITM), Image-grounded Text Generation(ITG) 세 가지 Loss를 활용하여 Q-former가 이미지를 바탕으로 텍스트를 만들어낼 수 있도록 학습

- * Step 2 : Freeze LLM pre-training

Step 1 에서 만들어진 모델을 LLM 과 FC layer 로 연결하여 다시 튜닝

Background **General-Purpose vision-language Model**

- BLIP-2 Training Strategy



- 1st step Pre-Training

Freeze Image encoder pre-training

- Image-text matching(ITM) 으로 supervised learning을 진행
- Image-text contrastive(ITC) pair 이미지와 텍스트는 가깝게, 서로 다른 이미지 텍스트는 similarity가 멀게 학습
- Image-grounded Text Generation(ITG)를 통해 이미지를 기반으로 텍스트를 생성하도록 학습

- 2nd step Pre-Training

Step 1 에서 만들어진 Q-former를 Frozen LLM 과 FC layer 로 연결하여 다시 튜닝

- Decoder based Model
: FC layer를 통해 Decoder의 토큰 임베딩으로 변환하여 연결시킨 후 pre-training
- Encoder-Decoder based Model
: FC layer를 통해 Encoder의 토큰 임베딩으로 변환 후 Prefix Text를 넣어 학습

➡ 결과적으로 Image-text-modality gap을 최소화하는 Q-former가 만들어짐

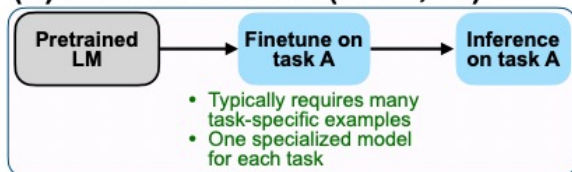
Background Instruction-Tuning

Instruction Tuning이 왜 필요하며 무엇일까?

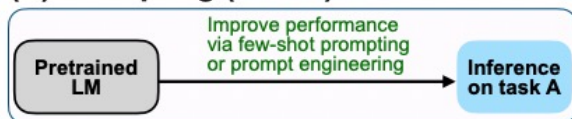
실제 학습데이터의 prompt 분포와 zero-shot을 할 때의 prompt 분포가 맞지 않는 문제가 발생

- Fine-tuning, Prompting, Instruction Tuning

(A) Pretrain-finetune (BERT, T5)



(B) Prompting (GPT-3)



(C) Instruction tuning (FLAN)

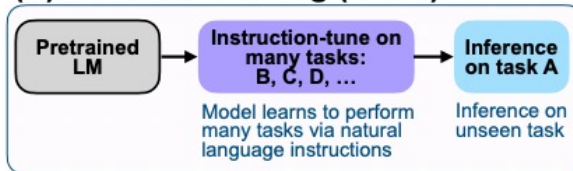
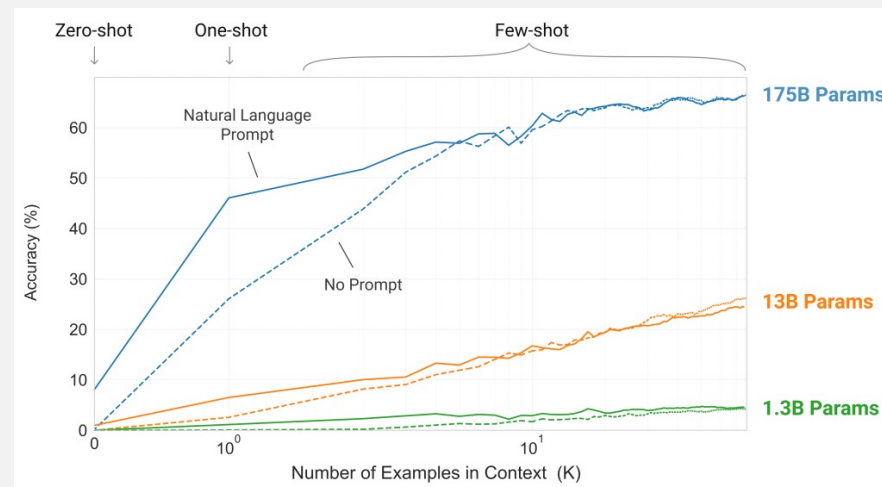


Figure 2: Comparing instruction tuning with pretrain-finetune and prompting.

- GPT-3 zero-shot, few-shot, prompting



- Instruction Tuning 이전에 개발된 LLM의 문제
 - Prompt 로 충분히 설명함에도 불구하고 Zero-shot 의 경우에 성능이 낮음
 - Reasoning과 같이 잘 못하는 Task도 존재함



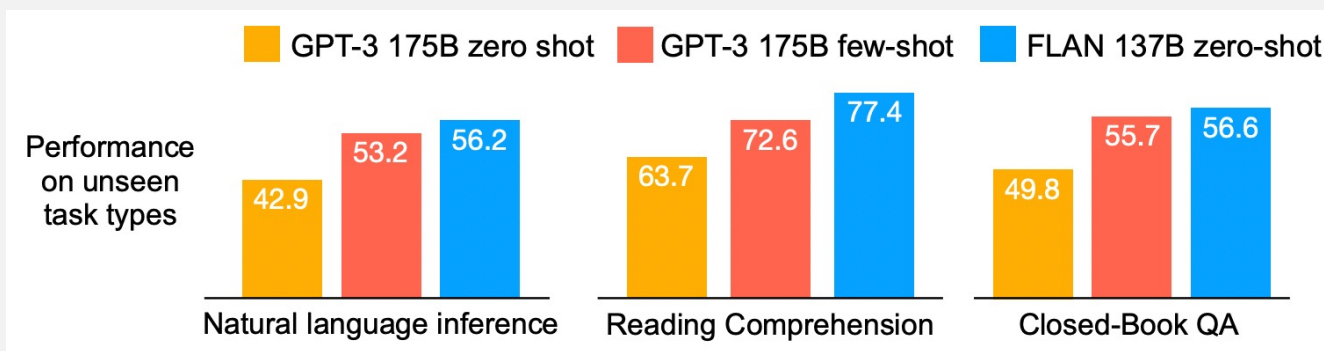
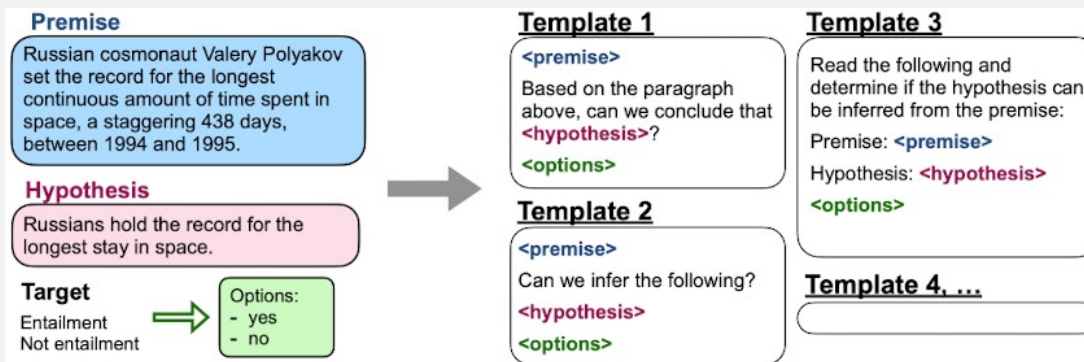
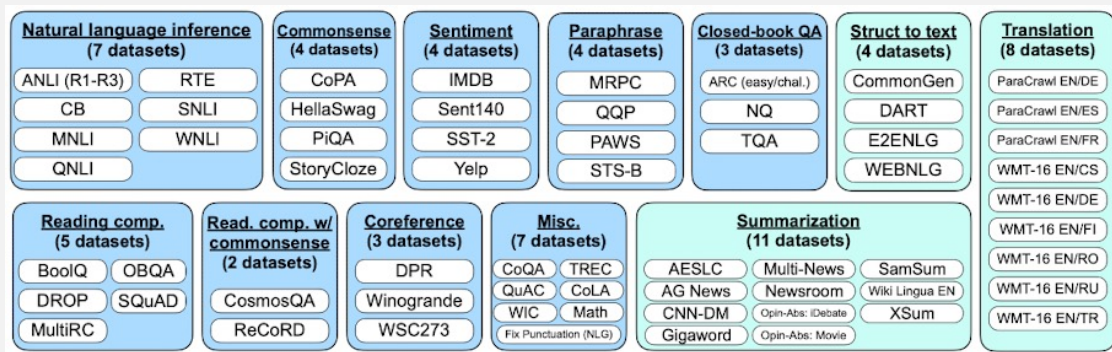
Prompt 로 충분한 설명을 함에도 Zero-shot의 성능이 낮은 문제를 데이터 관점에서 학습하자

Background Instruction-Tuning

- Instruction Tuning (FLAN)

Zero-shot 성능이 낮은 이유를 zero-shot prompt 형태가 학습된 prompt 형태와 다르다는 것을 원인으로 생각하여 실제 zero-shot prompt 형태로 다양한 지시문 template를 만들어 학습함

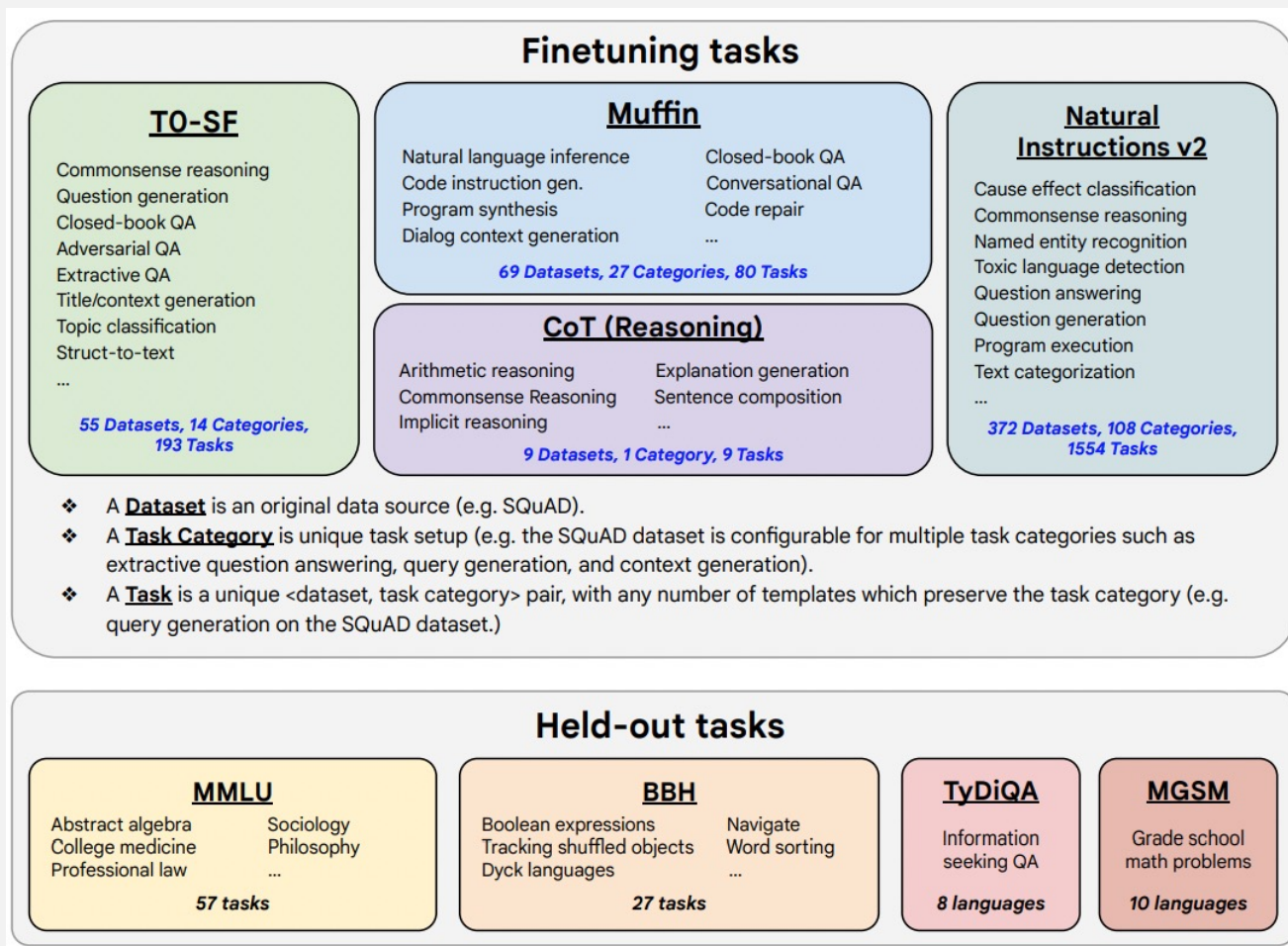
- 데이터를 12개 군집으로 분류하여 각 군집마다 템플릿을 생성함



➡ Instruction tuning 이후 zero-shot 성능이 크게 향상됨

Background Instruction-Tuning

- Flan-T5



Instruction을 생성하기 위해
비슷한 유형끼리 Cluster를 모았으며
Cluster 별로 Instruction template을 생성

- CoT : Chain-of-thought

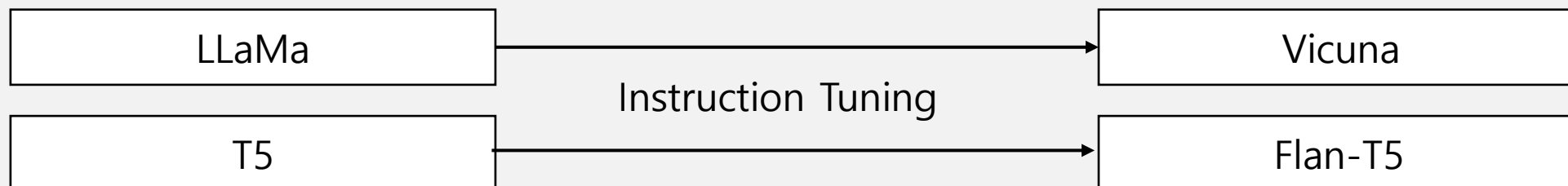
프롬프트 엔지니어링에서 사용하는 방법
Prompt를 수정하여 Step-by-step으로 결론에 도달하도록 제안

이를 Flan-T5 에서는 reasoning 데이터 셋으로 활용하여 구축
이를 활용했을 때 모든 evaluation성능 향상

Background LLM

- LLM 의 종류

LLM	학습 데이터	파라미터	아키텍처 구조	라이선스
OpenAI - GPT 3.5 (pre-trained)	499 GB	175 B	Decoder based	Commercial License
Facebook - LLaMA (pre-trained)	4.7 TB	7/13/33/65 B	Decoder based	Non-commercial bespoke license
LMSys - Vicuna (instruction-tuning)	No release	7/13 B	Decoder based	Non-commercial bespoke license
Google - Flan-T5 (instruction-tuning)	473 datasets, 146 task categories	80M~540B	Encoder-Decoder based	Apache license 2.0



Q & A

Introduction

- Definition problem

Vision-Language Task에서도 BLIP-2와 같이 frozen instruction-tuned LLM 이 나오면서 좋은 성과를 내고 있지만 아직까지도 다양한 Task에 대해 잘하기에는 부족하다.

- InstructBLIP이라는 vision-language instruction tuning framework를 제안
 - instruction tuning을 결합하여 **wide range visual tasks 성능 향상**
- Vision-language Instruction tuning 연구 분야에서 유의미하고 체계적인 정리를 함
 - 26 datasets 를 Instruction tuning format 으로 변환 11개의 task 카테고리 그룹핑
 - 13 개의 데이터는 **held-in datasets**으로써 **instruction tuning 에 활용**
 - 13 개의 데이터는 **held-out datasets**는 **zero-shot evaluation 에 활용**
- 본 논문의 저자는 instruction-aware visual feature extraction을 제안
 - Instruction이 단순히 LLM이 생성하는데 조건을 주는 것 뿐만 아니라 **image encoder에서 어떤 부분을 추출해야 하는지 조건을 줄 수 있도록 구성**
- 본 논문에서는 encoder-decoder based LLM인 FlanT5 와 decoder based 모델인 Vicuna 모델을 LLM 평가 모델로써 선정하였으며 **다양한 데이터 셋에 대해 SOTA 달성**

Can you describe this image in detail?



The image depicts a person standing in front of an open doorway, looking out into space. They are wearing a jacket and appears to be contemplating the vastness of the universe beyond the doorway. In the background, there are multiple stars visible, creating a stunning celestial scene. The person's gaze is directed towards the center of the image, where a bright star can be seen shining brightly. There are also smaller stars scattered throughout the scene, adding to the overall cosmic atmosphere.

What if they jump over the doorway?



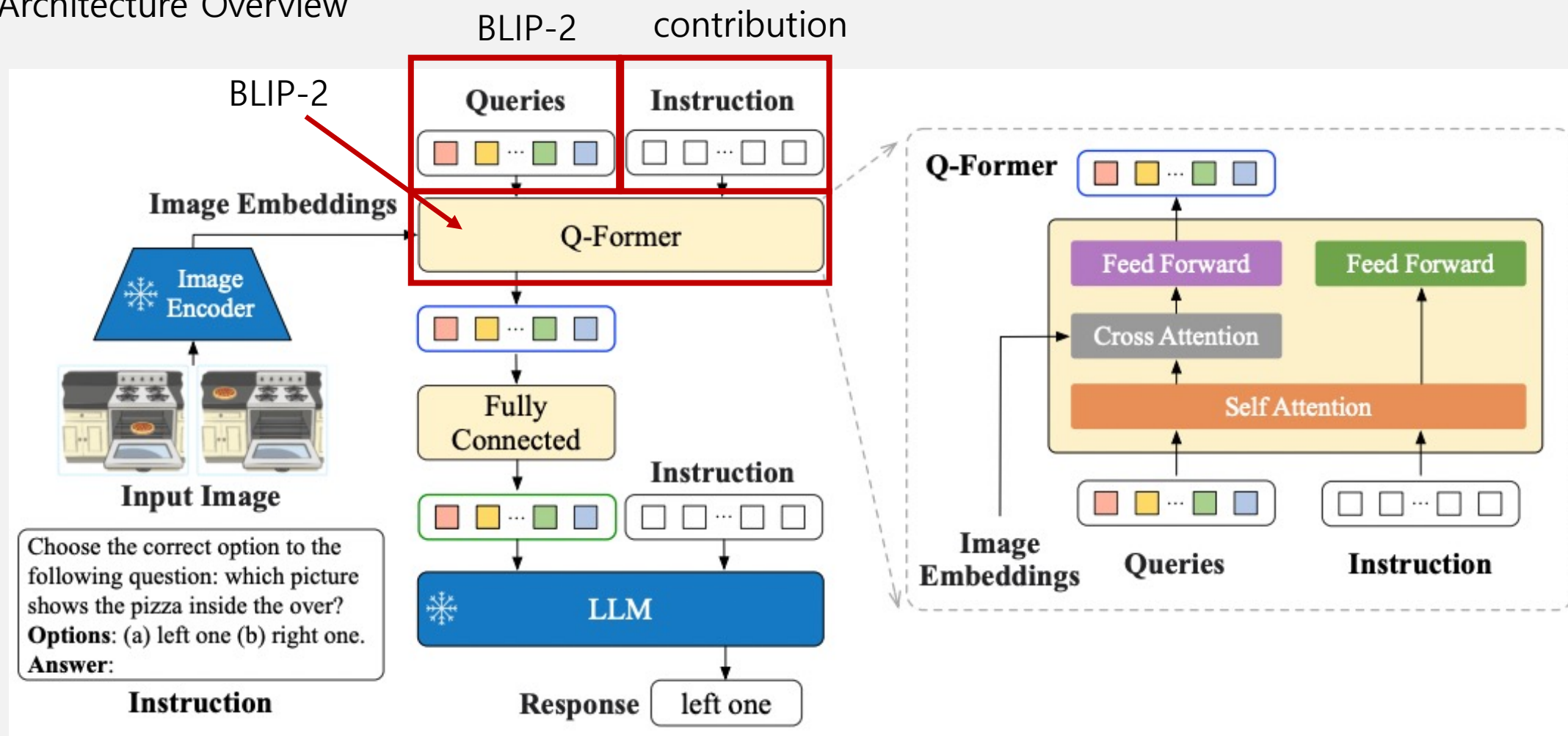
If the person were to jump over the doorway, they would be leaving the confines of their current location and venturing into the vastness of space. This could be a metaphorical representation of stepping out of one's comfort zone or embarking on a new adventure. However, it is important to note that such an action would be extremely dangerous and potentially life-threatening due to the harsh conditions of space.

Vision-Language Instruction Tuning

- Tasks and Datasets
- Training and Evaluation Protocols
- Instruction-aware Visual Feature Extraction
- Training Datasets Balancing
- Inference Methods
- Implementation Details

Vision-Language Instruction Tuning Tasks and Datasets

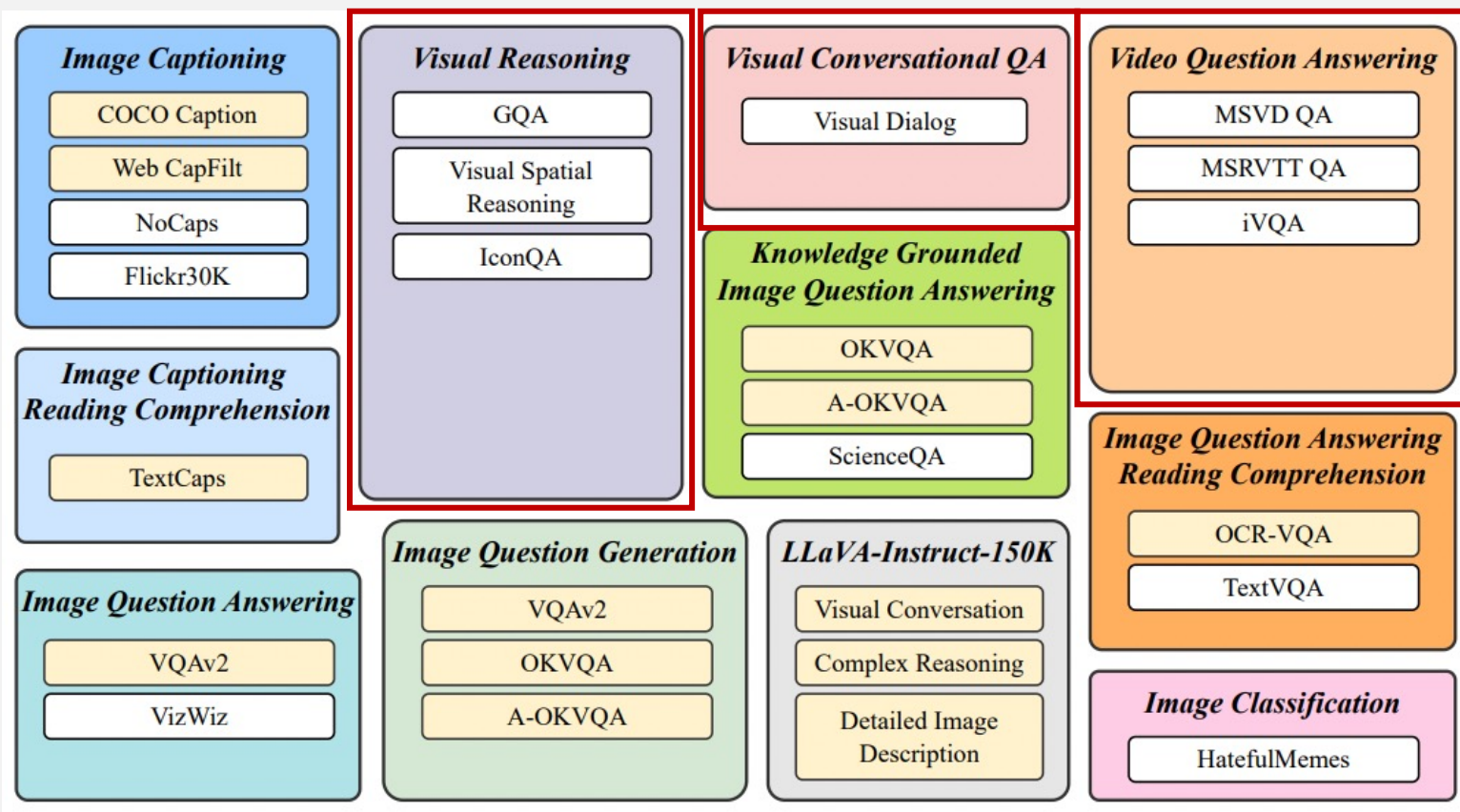
- Architecture Overview



Vision-Language Instruction Tuning Tasks and Datasets

- Tasks and Datasets

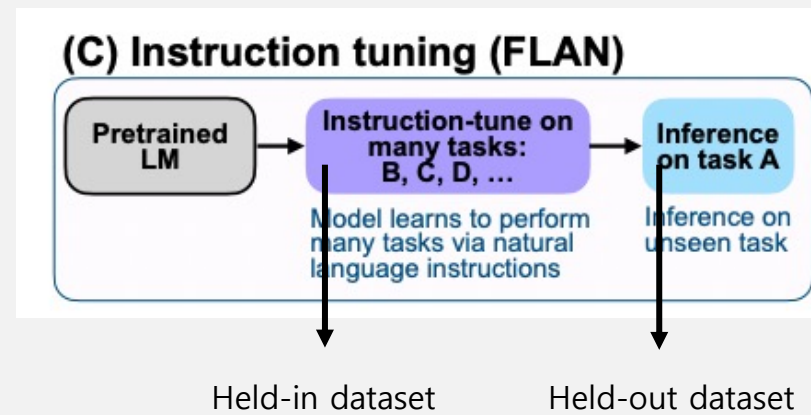
□ : Zero-shot evaluation 용



노란색 : Instruction Tuning용 데이터셋 (held-in)

흰색 : Evaluation 을 위한 데이터셋 (held-out)

데이터셋을 11가지 클러스터로 분류하였으며 Held-in, Held-out Dataset으로 분류하여 Instruction Tuning, Evaluation 진행



Vision-Language Instruction Tuning Tasks and Datasets

- Instruction for Instruction Tuning

Task	Instruction Template
Image Captioning	<code><Image></code>A short image caption: <code><Image></code>A short image description: <code><Image></code>A photo of <code><Image></code>An image that shows <code><Image></code>Write a short description for the image. <code><Image></code>Write a description for the photo. <code><Image></code>Provide a description of what is presented in the photo. <code><Image></code>Briefly describe the content of the image. <code><Image></code>Can you briefly explain what you see in the image? <code><Image></code>Could you use a few words to describe what you perceive in the photo? <code><Image></code>Please provide a short depiction of the picture. <code><Image></code>Using language, provide a short account of the image. <code><Image></code>Use a few words to illustrate what is happening in the picture.
VQA	<code><Image></code>{Question} <code><Image></code>Question: {Question} <code><Image></code>{Question} A short answer to the question is <code><Image></code>Q: {Question} A: <code><Image></code>Question: {Question} Short answer: <code><Image></code>Given the image, answer the following question with no more than three words. {Question} <code><Image></code>Based on the image, respond to this question with a short answer: {Question}. Answer: <code><Image></code>Use the provided image to answer the question: {Question} Provide your answer as short as possible: <code><Image></code>What is the answer to the following question? "{Question}" <code><Image></code>The question "{Question}" can be answered using the image. A short answer is
VQG	<code><Image></code>Given the image, generate a question whose answer is: {Answer}. Question: <code><Image></code>Based on the image, provide a question with the answer: {Answer}. Question: <code><Image></code>Given the visual representation, create a question for which the answer is "{Answer}". <code><Image></code>From the image provided, craft a question that leads to the reply: {Answer}. Question: <code><Image></code>Considering the picture, come up with a question where the answer is: {Answer}. <code><Image></code>Taking the image into account, generate an question that has the answer: {Answer}. Question:

- Instruction for zero-shot evaluation

GQA, VizWiz, iVQA, MSVD, MSRVT `<Image>` Question: {} Short answer:

NoCaps, Flickr30k `<Image>` A short image description:

TextVQA `<Image>` OCR tokens: {}. Question: {} Short answer:

IconQA `<Image>` Question: {} Options: {}. Short answer:

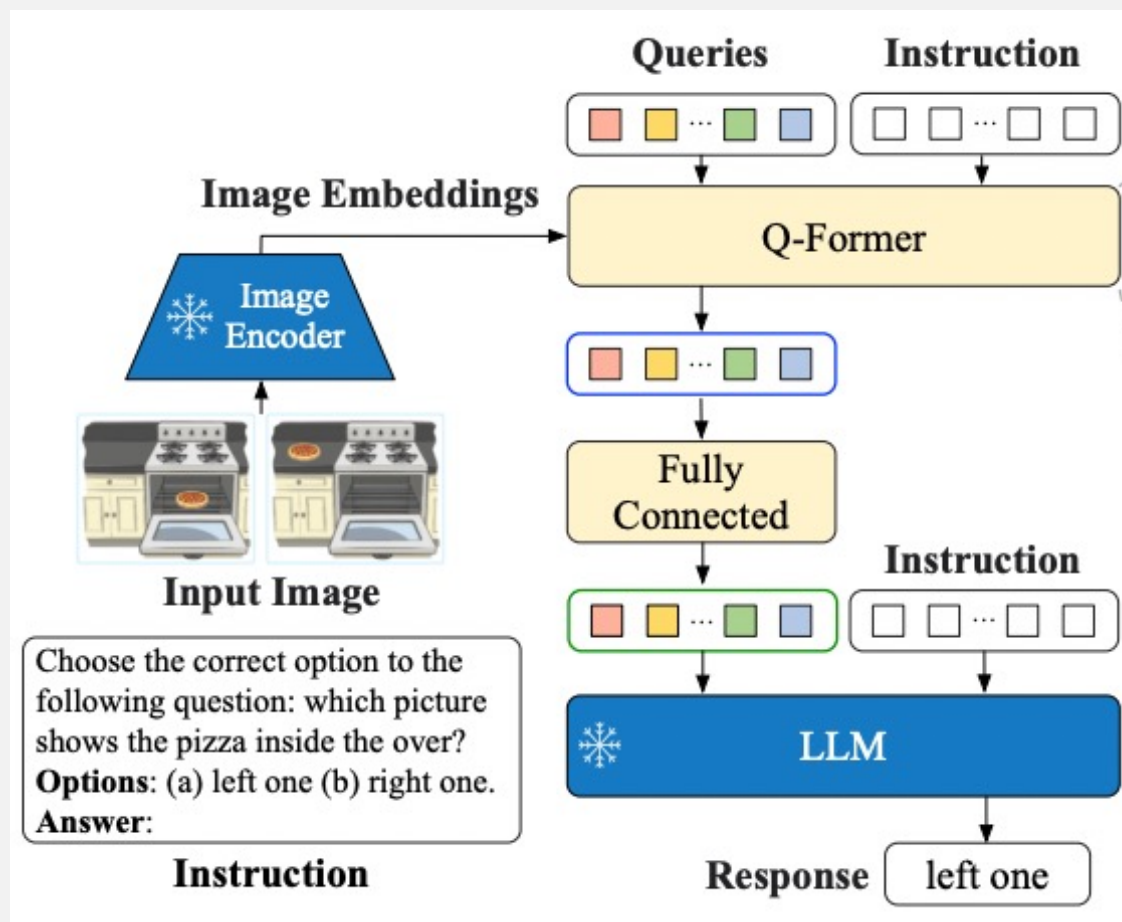
ScienceQA `<Image>` Context: {} Question: {} Options: {}. Answer:

HatefulMemes `<Image>` This is an image with: "{}" written on it. Is it hateful? Answer:

VSR `<Image>` Based on the image, is this statement true or false? "{}" Answer:

Visual Dialog `<Image>` Dialog history: {} \n Question: {} Short answer:

Vision-Language Instruction Tuning Instruction-aware Visual Feature Extraction



- 이전의 Image-to-Text generation

Instruction-agnostic method

실제 사용하는 Instruction을 고려하지 않음

가정 : Task마다 Instruction을 다르게 주어 image feature representation 을 뽑는 방식을 적절하게 조정 하는 것이 더 좋음



- InstructBLIP의 instruction-aware Q-former

Instruction-aware method

Instruction을 바탕으로 Visual Representation 을 적절하게 추출함

결과적으로 같은 이미지가 들어오더라도 서로 다른 테스크라면 다르게 동작을 하게 됨

Vision-Language Instruction Tuning Balancing Training Datasets

- 데이터셋마다 크기가 매우다른데 이로 인해 smaller dataset에는 Overfitting이 일어나고 많은 데이터셋에는 Underfitting이 일어나는 문제점을 해소하고자 함
- 데이터셋마다 확률적으로 추출하는데 큰 데이터셋은 많이 줄어들과 작은 데이터셋은 조금 줄어들도록 샘플링

$$p_d = \frac{\sqrt{S_d}}{\sum_{i=1}^D \sqrt{S_i}}$$

이 샘플링 식은 실험적으로 알게된 공식

- 학습하는 데이터의 난이도에 따른 weight 변경
A-OKVQA와 같이 feature multi-choice-question같이 쉬운 문제는 weight를 줄여서 학습
OKVQA 와 같이 open-ended text generation같은 경우에는 weight를 늘려서 학습



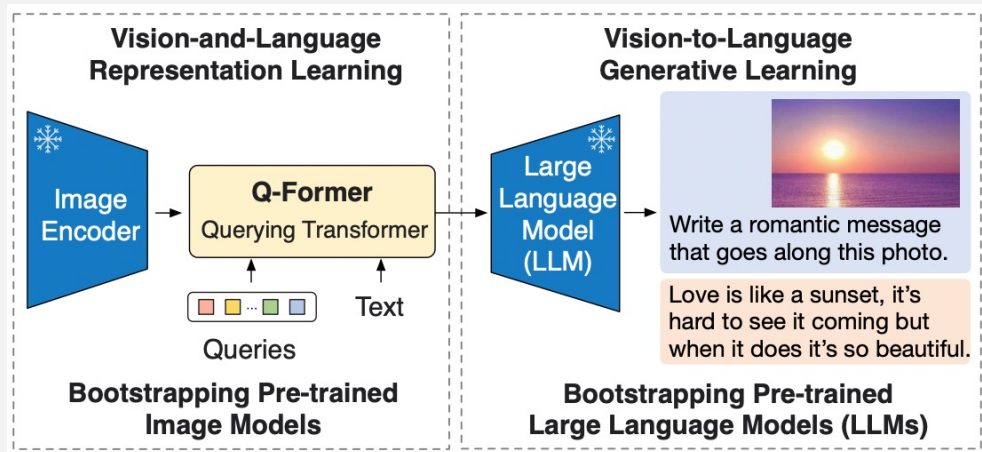
이러한 학습 전략을 통해 held-in, held-out generalization 성능이 향상됐다.

Vision-Language Instruction Tuning Inference Methods

- 데이터셋마다 조금씩 다른 generation approach를 적용함
- Image Captioning, Open-ended VQA
instruction-tuned model이 바로 Prompt를 받고 response를 내뱉은 후에 정답과 얼마나 다른지 evaluation
- Classification, Multi-choice VQA
classification이나 multi-choice VQA의 경우에는 ranking으로 평가
프롬프트로 정답을 생성해내도록 하되 vocabulary는 candidates 리스트에서 추출하도록 함
이후 log-likelihood를 각각의 후보군과 선택한 것 중에 가장 높은것을 최종 예측값으로 선정
- Binary classification
positive와 negative를 의미하는 것은 자주 나오는 자연스러운 단어로 확장시킴
Positive는 yes, true 와 같이 나왔을 때 맞은 것으로 평가하였고 Negative는 no, false 로 확장시켜 평가함

Vision-Language Instruction Tuning Implementation Details

- Architecture



- Model Weight

Pre-trained BLIP-2 checkpoints 활용

Pre-trained Q-former, Pre-trained FlanT5

Vicuna 모델은 BLIP-2 에서 튜닝한 모델이 없어 그대로 활용

- Image Encoder

BLIP-2에서 image-encoder로 쓴 ViT-g/14를 활용

- Text Decoder

LLM으로는 FlanT5-XL (3B), FLanT5-XXL (11B), Vicuna-7B, Vicuna-13B을 활용함

- Training & Hyper parameters

FlanT5 : Instruction-tuned T5, Vicuna : Instruction-tuned LLaMA

Package	Batch size	Training step	Optimizer	scheduler	Training duration
LAVIS	192/128/64 3B/7B/11,13B	60K steps 3K validation	AdamW B1 = 0.9, B2=0.999 weight decay=0.05	Cosine warmup scheduler 1000 step 10-8~10-5 warm-up Cosine decay to 0	1.5 days 16 A100(40G)

Q & A

Experimental Results and Analysis

Experimental Results and Analysis zero-shot Evaluation

- Held-out evaluation

	NoCaps	Flickr 30K	GQA	VSR	IconQA	TextVQA	Visdial	HM	VizWiz	SciQA image	MSVD QA	MSRVTT QA	iVQA
Flamingo-3B [4]	-	60.6	-	-	-	30.1	-	53.7	28.9	-	27.5	11.0	32.7
Flamingo-9B [4]	-	61.5	-	-	-	31.8	-	57.0	28.8	-	30.2	13.7	35.2
Flamingo-80B [4]	-	67.2	-	-	-	35.0	-	46.4	31.6	-	35.6	17.4	40.7
BLIP-2 (FlanT5 _{XL}) [20]	104.5	76.1	44.0	60.5	45.5	43.1	45.7	53.0	29.8	54.9	33.7	16.2	40.4
BLIP-2 (FlanT5 _{XXL}) [20]	98.4	73.7	44.6	68.2	45.4	44.1	46.9	52.0	29.4	64.5	34.4	17.4	45.8
BLIP-2 (Vicuna-7B)	107.5	74.9	38.6	50.0	39.7	40.1	44.9	50.6	25.3	53.8	18.3	9.2	27.5
BLIP-2 (Vicuna-13B)	103.9	71.6	41.0	50.9	40.6	42.5	45.1	53.7	19.6	61.0	20.3	10.3	23.5
InstructBLIP (FlanT5 _{XL})	119.9	84.5	48.4	64.8	50.0	46.6	46.6	56.6	32.7	70.4	43.4	25.0	53.1
InstructBLIP (FlanT5 _{XXL})	120.0	83.5	47.9	65.6	51.2	46.6	48.5	54.1	30.9	70.6	44.3	25.6	53.8
InstructBLIP (Vicuna-7B)	123.1	82.4	49.2	54.3	43.1	50.1	45.2	59.6	34.5	60.5	41.8	22.1	52.2
InstructBLIP (Vicuna-13B)	121.9	82.8	49.5	52.1	44.8	50.7	45.4	57.5	33.4	63.1	41.2	24.8	51.0

Experimental Results and Analysis Ablation Study on Instruction Tuning Techniques

- Instruction-aware Visual Features, Data Balancing Experiments

Model	Held-in Avg.	GQA	ScienceQA (image-context)	IconQA	VizWiz	iVQA
InstructBLIP (FlanT5 _{XL})	94.1	48.4	70.4	50.0	32.7	53.1
w/o Instruction-aware Visual Features	89.8	45.9 (↓2.5)	63.4 (↓7.0)	45.8 (↓4.2)	25.1 (↓7.6)	47.5 (↓5.6)
w/o Data Balancing	92.6	46.8 (↓1.6)	66.0 (↓4.4)	49.9 (↓0.1)	31.8 (↓0.9)	51.1 (↓2.0)
InstructBLIP (Vicuna-7B)	100.8	49.2	60.5	43.1	34.5	52.2
w/o Instruction-aware Visual Features	98.9	48.2 (↓1.0)	55.2 (↓5.3)	41.2 (↓1.9)	32.4 (↓2.1)	36.8 (↓15.4)
w/o Data Balancing	98.8	47.8 (↓1.4)	59.4 (↓1.1)	43.5 (↑0.4)	32.3 (↓2.2)	50.3 (↓1.9)

Table 2: Results of ablation studies that remove the instruction-aware Visual Features (Section 2.3) and the balanced data sampling strategy (Section 2.4). For held-in evaluation, we compute the average score of four datasets, including COCO Caption, OKVQA, A-OKVQA, and TextCaps. For held-out evaluation, we show five datasets from different tasks.

Experimental Results and Analysis Qualitative Evaluation

Introduce me this painting
in detail.



The painting depicts a young girl with a pearl earring, also known as "The Girl with the Pearl Earring" or "Girl with a Pearl Earring". She is wearing a blue dress and has a pearl earring on her left ear. Her hair is pulled back into a bun, and she is looking directly at the viewer. This painting is a famous work of art by Dutch painter Johannes Vermeer, created in 1665. It is considered one of his masterpieces and is widely recognized for its intricate details and realistic portrayal of the young girl's facial expression.



Scene을 보거나 이미지 내의 특징을 잡는것 모두 잘함



What could have happened based on the
current scene?



Based on the current scene in the image, it is possible that a hurricane or severe weather event caused significant damage to the buildings and infrastructure in the area. The person walking through the debris-covered area suggests that they are inspecting the damage caused by the disaster. The presence of palm trees indicates that the location may be in a tropical or sub-tropical region, which can be more susceptible to hurricanes and other natural disasters.

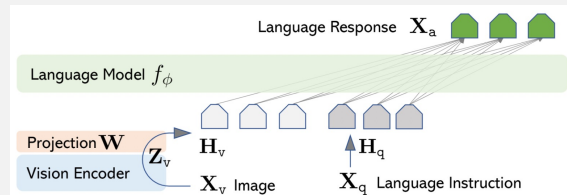
Experimental Results and Analysis Qualitative Evaluation

LLaVA, MiniGPT-4, ChatGPT-4 모델과 비교

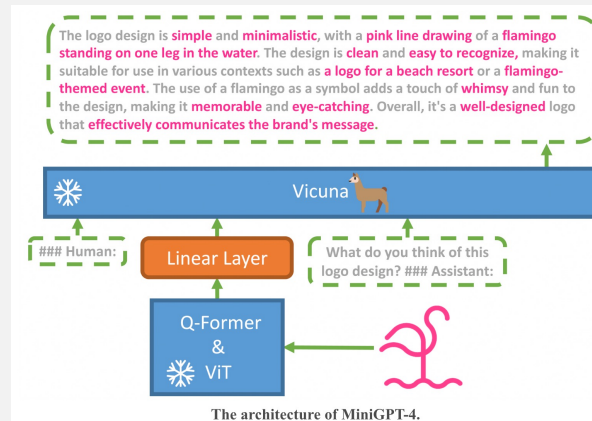
LLaVA, MiniGPT-4는 길고 관련 없는 생성 패턴을 보임

InstructBLIP은 길지만 적절한 내용을 추출하여 답변함

* LLaVA



* MiniGPT-4



[Input Image]



[Input Instruction] What is unusual about this image?

[Model Response]

InstructBLIP



The unusual aspect of this image is that a man is ironing clothes on the back of a yellow taxi cab while it is driving down a busy city street. He is standing on top of the car's trunk, holding an iron and using it to press his clothing items. This situation is not only unconventional but also potentially hazardous, as the man could be at risk of falling off the vehicle or causing an accident due to his actions.

GPT-4



The unusual thing about this image is that a man is ironing clothes on an ironing board attached to the roof of a moving taxi.

LLaVA

The unusual aspect of this image is a man ironing clothes on the back of a minivan or van. This is not a typical place to perform this activity, as one would usually iron clothes in a more stationary and safe location, such as a home, using a regular ironing board. The scene depicted in the image is peculiar as it involves a makeshift ironing setup on a vehicle, which can be both unsafe and unconventional. Additionally, it is not clear how the man is able to maintain balance and stability while ironing clothes in such an unstable environment.

MiniGPT-4

The image shows a man standing on the back of a yellow taxi, with a yellow shirt and black pants, and a blue backpack on his back. The taxi is driving on a city street with other cars and taxis in the background.

Experimental Results and Analysis Instruction Tuning vs. Multitask Learning

- Multi-task learning, BLIP-2 Zero-shot, InstructBLIP Zero-shot

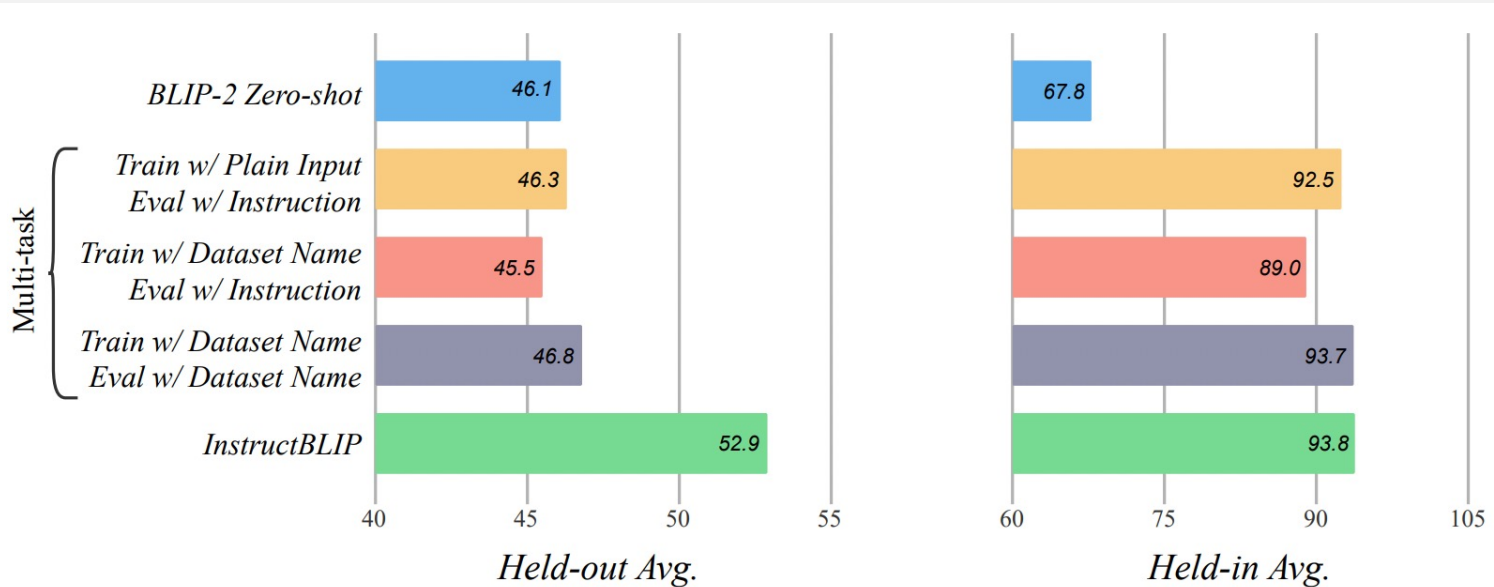


Figure 4: Comparison of instruction tuning and multitask training based on BLIP-2 FlanT5_{XL} backbone. For held-in evaluation, we compute the average score across all held-in datasets. For held-out evaluation, we compute the average score across GQA, TextVQA, VSR, HatefulMemes, IconQA, ScienceQA, iVQA, VizWiz.

- Multi-task learning

1st approach

Training : 데이터셋의 input-output format

Evaluation : instruction을 넣어 평가

* Image captioning만 instruction을 뺄 때
성능이 더 좋았음

2nd approach

Training : task와 dataset을 지정하여 학습

Evaluation : task와 dataset을 지정하여 평가

ex) [Visual question answering:VQAv2]

(held-out dataset의 경우에는 dataset 명
을 넣지 않고 task만 넣어줌)



InstructBLIP 방식이 Held-in dataset에서는 Multi-task learning에 비해 성능향상이 미약했지만
Held-in dataset과 held-out dataset에도 모두 성능향상이 됨

Experimental Results and Analysis Finetuning InstructBLIP on Downstream Tasks

- Flamingo와 BLIP-2에서 했던것과 마찬가지로 Downstream tasks를 적용

	ScienceQA IMG	OCR-VQA	OKVQA	A-OKVQA			
				Direct Answer Val	Test	Multi-choice Val	Test
Previous SOTA	LLaVA [25] 89.0	GIT [43] 70.3	PaLM-E(562B) [9] 66.1	[15] 56.3	[37] 61.6	[15] 73.2	[37] 73.6
BLIP-2 (FlanT5 _{XXL})	89.5	72.7	54.7	57.6	53.7	80.2	76.2
InstructBLIP (FlanT5 _{XXL})	90.7	73.3	55.5	57.1	54.8	81.0	76.7
BLIP-2 (Vicuna-7B)	77.3	69.1	59.3	60.0	58.7	72.1	69.0
InstructBLIP (Vicuna-7B)	79.5	72.8	62.1	64.0	62.1	75.7	73.4

Table 3: Results of finetuning BLIP-2 and InstructBLIP on downstream datasets. Compared to BLIP-2, InstructBLIP provides a better weight initialization model and achieves SOTA performance on three out of four datasets.

 Flan-T5는 multi-choice tasks를 잘하고 Vicuna는 open-ended generation tasks를 잘하는 것을 발견

Related Work

- NLP Instruction Tuning
instruction template을 구축하여 Instruction tuning을 적용하고 generalization 성능을 올림
- Datasets
 - NLP
GPT-3 와 같은 LLM 으로 데이터를 구축하기도 함
 - Vision-Language Model
GPT-4 로 Image Captioning 데이터를 구축하기도 함 (LLaVA)
- Vision-Language Models
BLIP-2 에서 frozen FlanT5, OPT 모델, Q-former를 활용
MiniGPT-4 에서는 BLIP-2의 encoder, Q-former를 Vicuna와 Linear 모듈로 연결하고 Linear layer만 학습
mPLUG-owl 은 LLaMA를 LoRA를 활용하여 학습
MultilInstruct 는 vision-language instruction tuning에서 Pre-trained LLM을 활용하지 않아 성능이 안좋았음

1. InstructBLIP은 wider range vision-language instruction data를 활용하여 template based Converted data를 활용
2. Pre-trained LLM을 활용
3. instruction-aware visual feature extraction Mechanism을 제안
4. Unseen tasks에도 generalization 성능을 높이는데 이점이 있었음

Conclusion

- 간단하고 기본적인 Instruction Tuning framework를 vision-language models로 제안함
- Unseen tasks에 대해 generalization 성능을 올렸으며 wide-range unseen Task에서 SOTA를 달성
- Qualitative Examples을 봤을 때 instruction following 성능을 보여줌
(complex visual reasoning, knowledge-grounded image description, and multi-turn conversations)
- 심지어 다양한 downstream tasks에 대해서도 SOTA를 달성함

Q & A
